R-SOFT

PRO_B
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PRo.com

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1. EXECUTIVE SUMMARY

This Business Requirements Document (BRD) presents the analysis and proposed solution for improving attendance tracking and event management within a local airsoft team through the development of the **R-Soft mobile application**.
Currently, attendance is managed manually using spreadsheets, resulting in inefficiencies, delays during events, and increased risk of human error. These limitations negatively impact the team's ability to follow event schedules and maintain accurate participation records.

The proposed solution introduces a centralized mobile platform that automates check-in processes using QR codes, provides real-time event information, and improves communication among stakeholders.

The intended audience of this document includes **Team Leaders, Players, and Field Owners**, all of whom will directly benefit from improved coordination, accuracy, and operational efficiency.

2. PROJECT OBJECTIVES

The primary objective of this project is to design and implement a mobile application that replaces manual attendance tracking with an automated and efficient system.

The current process relies heavily on manual data entry, which is time-consuming and prone to inconsistencies. Additionally, there is limited visibility into participant data during events.

The project aims to achieve the following objectives:

- Reduce manual check-in time by approximately **90%**
- Increase event participation by **20% within the first year**
- Improve overall event organization and coordination
- Enhance data accuracy and accessibility

3. PROJECT SCOPE

The R-Soft project focuses on the development of a mobile application designed to improve the management of airsoft events and automate attendance tracking processes. The system aims to replace manual spreadsheet-based tracking with a centralized, digital solution that enhances efficiency, accuracy, and coordination among all participants.

At a high level, the project includes the design and implementation of core functionalities such as QR code-based check-in, event creation and scheduling, team roster management, and real-time attendance tracking. The system will also provide notification features, basic reporting

capabilities, and secure storage of user and emergency contact information. Additionally, the application will be developed as a cross-platform solution compatible with both Android and iOS devices and will support offline functionality with data synchronization once connectivity is restored.

The scope is intentionally limited to essential features required for efficient event and attendance management. Advanced functionalities such as e-commerce features, user-to-user messaging, and GPS tracking are excluded from the initial version in order to maintain simplicity, reduce development complexity, and stay within budget constraints.

Overall, the project scope is defined to deliver a practical, low-cost, and scalable solution that addresses the primary operational challenges of airsoft teams while allowing for future expansion and feature enhancements.

IN SCOPE

The following features are included within the scope of the project:

- *QR code-based check-in system*
- *Event calendar and management*
- *Team roster and role management*
- *Attendance tracking and history*
- *Notifications and reminders*
- *Basic reporting (attendance, trends)*

OUT OF SCOPE

The following features are excluded from the initial release:

- *Selling airsoft gear or consumables*
- *User-to-user messaging*
- *GPS tracking*

4. BUSINESS DRIVERS

BUSINESS DRIVER #1 – Operational Efficiency

The current manual system results in wasted time and inefficiencies during events. Automating attendance tracking significantly improves workflow and reduces administrative burden.

BUSINESS DRIVER #2 – Cost Effectiveness

By reducing manual administrative work, the system provides measurable cost savings and efficient use of time resources.

BUSINESS DRIVER #3 – Improved Coordination

Centralizing event information enhances communication, reduces confusion, and increases participation among players.

5. PROPOSED PROCESSES

The proposed process introduces a centralized mobile application (R-Soft) designed to replace the current manual attendance tracking system with an automated and efficient digital solution. The system streamlines event management, participant registration, and attendance tracking through an integrated and user-friendly workflow.

Initially, the Team Leader creates an event within the application by specifying key details such as date, time, location, and participant limits. Once the event is created, it becomes visible to all registered users, who can view event details and confirm their participation directly through the application.

Prior to the event, users receive notifications and reminders, ensuring improved awareness and participation. Upon arrival at the event location, participants complete the check-in process by scanning a QR code generated specifically for that event. This process allows attendance to be recorded instantly and accurately, eliminating the need for manual input.

To address connectivity challenges commonly encountered in remote outdoor environments, the system supports offline functionality. During periods without internet access, attendance data and user actions are stored locally on the device. Once a stable connection is re-established, the system automatically synchronizes all stored data with the central database, ensuring consistency and reliability.

Throughout the event, the Team Leader has access to real-time attendance information, enabling better coordination and decision-making. After the event, the system stores participation data, allowing users to view their attendance history and enabling basic reporting and analysis.

A visual representation of the proposed process flow and user interface is provided in **Appendix** , including wireframes illustrate the system interactions.

6. FUNCTIONAL REQUIREMENTS

PRIORITY

Value	Rating	Description
1	Critical	The requirement is critical to the project's success. Without fulfilling this requirement, the project is not possible.
2	High	The requirement is high priority re the project's success, but the project could still be implemented in a minimum viable product (MVP) scenario.
3	Medium	The requirement is important to the project's success, as it provides value, but the project could still be implemented in an MVP scenario.
4	Low	The requirement is low priority (i.e., it would be nice to have), but the project's success is not dependent upon it.
5	Future	The requirement is outside of the project's scope and is included as a possible component of a prospective release and/or feature.

REQUIREMENTS CATEGORIES (RC1)

ID	Requirement	Priority	Raised By
RC 1	QR code check-in system	1	Team Leader
RC 2	Event creation & management	1	Team Leader
RC 3	Offline functionality with sync	1	Developer
RC 4	Emergency contact storage	1	Safety Requirement
RC 5	User-friendly interface	1	All stakeholders
RC 6	Secure authentication & data protection	1	System
RC 7	Role-based access control	2	Admin
RC 8	Real-time participant updates	2	Players
RC 9	Participation modification/cancel	2	Players
RC 10	Notifications & reminders	3	Players
RC 11	Attendance history tracking	3	Team Leader
RC 12	Reporting & analytics	3	Team Leader

7. USER STORIES (AGILE MAPPING OF REQUIREMENTS)

• As a player, I want to see upcoming games for the season so that I can plan my attendance.

• As an event planner, I want to know how many participants are attending an event so that I can organize it effectively.

• As a field owner, I want to know how many vehicles are expected so that I can prepare sufficient parking spaces.

• As a player, I want to see who is attending an event so that I can know if my friends will be there.

8. NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
NFR 1	System must work offline with later synchronization
NFR 2	High performance during events (fast QR scanning)
NFR 3	Secure data storage and authentication
NFR 4	Cross-platform (Android & iOS)
NFR 5	Scalable architecture for future growth
NFR 6	Usable in outdoor/high-glare environments
NFR 7	Compatible with mobile devices with lower processing power
NFR 8	Intuitive and easy to use interface
NFR 9	Reliable real-time updates when online

9. PROJECT CONSTRAINTS

• **Budget:** The project is self-funded with a total budget of $0. This means all development must rely on free tools, open-source frameworks, and free-tier services. No paid APIs or commercial infrastructure can be used, which requires efficient design choices and careful resource management throughout the project.

• **Technology Compatibility**: The system must be developed as a cross-platform application compatible with both Android and iOS devices. This ensures that all users can access the platform regardless of their mobile operating system, preventing exclusion and maintaining consistency across devices.

• **Environment**: The application must be fully usable in outdoor environments where lighting conditions may vary significantly. In particular, the interface must remain readable in high-glare situations such as direct sunlight, requiring a high-contrast UI design.

• **Offline Data Synchronizatio**n: Airsoft games are frequently conducted in remote areas where network connectivity is unreliable or completely unavailable. As a result, requiring a constant internet connection would lead to system failures during critical operations such as

check-ins and event management, forcing users to revert to manual, paper-based tracking methods, which significantly reduces efficiency and accuracy. To address this issue, the system will implement local data storage combined with asynchronous background synchronization. This approach will allow users to continue using the application seamlessly in offline mode, with all data stored locally on the device and automatically synchronized with the central system once a stable internet connection is restored.

- **Security**: The system must ensure that all user data, including personal information and emergency contact details, is securely stored and protected. Authentication mechanisms must be implemented to prevent unauthorized access, while still relying on free or open-source security solutions due to budget limitations.
- **Performance**: The application must remain responsive and efficient during normal usage, even when handling multiple users, events, and check-ins simultaneously. Core functions such as event loading, attendance updates, and QR scanning must have minimal delay to ensure smooth operation during live events.
- **Scalability**: Although initially designed for small airsoft communities, the system must be structured to allow future growth. This includes the ability to support additional users and event locations without requiring major architectural changes.
- **Device Hardware**: The application must be optimized to run on a wide range of mobile devices, including older smartphones with limited processing power, memory, and storage capacity. This ensures accessibility for all users regardless of their hardware specifications.
- **Usability**: The system must be simple and intuitive to use, requiring minimal training or technical experience. This is particularly important for fast-paced environments where users need to quickly check in, view events, or manage participation.
- **Connectivity**: The system must not depend on constant internet access for essential features such as event viewing, check-ins, and roster access. Offline functionality must be fully supported, with automatic data synchronization when connectivity is restored.

10. COST AND BENEFIT

The proposed system is designed to be a **low-cost**, **self-funded solution**, with overall costs primarily consisting of development time and an estimated $20 per month for server hosting and backend infrastructure. Additional costs include the initial effort required to implement features such as offline functionality, QR code integration, and cross-platform compatibility for both Android and iOS devices. Ongoing maintenance, including bug fixes, updates, and performance improvements, may also require a small but continuous time investment from the developer.

Despite these relatively low costs, **the benefits of the system are significant**. The application

automates key administrative tasks such as attendance tracking, event management, and player coordination, reducing the workload of the Team Leader by an estimated two hours per week. When valuing this time at approximately $20 per hour, the system effectively recovers its operational cost within the first week of use. In addition, improved organisation, reduced manual errors, and increased efficiency in event coordination provide ongoing value to all stakeholders, including players and field owners. The inclusion of offline functionality and real-time updates further ensures reliable performance in all environments, enhancing the overall usability and dependability of the system.
Overall, the long-term operational benefits and time savings significantly outweigh the minimal financial costs, making the system a cost-effective and practical solution for managing airsoft events.

11. RESOURCES

The successful development and implementation of the R-Soft application require a combination of human, technical, and operational resources. Given the self-funded nature of the project, all resources are selected to minimize costs while maintaining efficiency and quality.

- **Personnel Resources:** The project will primarily be developed and managed by a single developer, responsible for system design, implementation, testing, and deployment. Additional support will be provided by a small group of volunteer users (airsoft players) who will participate in testing activities and provide feedback for system improvements.
- **Hardware Resources:** The project requires access to standard computing equipment, including a personal computer or laptop for development purposes. Additionally, multiple mobile devices (smartphones) running Android and iOS operating systems will be needed for testing to ensure cross-platform compatibility and performance across different hardware configurations.
- **Software Resources:** Development will rely on free and open-source tools and frameworks. These may include a cross-platform mobile development framework, code editors or integrated development environments (IDEs), version control systems, and database management solutions. Cloud-based backend services and hosting platforms will be utilized through free-tier or low-cost plans to support data storage and synchronization.
- **Equipment and Tools:** No specialized physical equipment is required beyond standard development tools. QR code generation and scanning functionalities will be implemented using software libraries integrated into the application.
- **Infrastructure and Office Space:** The project will be developed in a remote or home-based environment, eliminating the need for dedicated office space. Communication and

coordination (if needed) will be conducted through online collaboration tools.

• **Testing Resources:** Testing will be conducted in real-world conditions during airsoft events to validate system performance, particularly in offline scenarios and outdoor environments. Feedback collected during these sessions will be used to refine and improve the application.

12. SCHEDULE

Phase	Duration	Key Outcome
Planning	1–2 weeks	Approved requirements
Design	1 week	System design completed
Development	4 weeks	Functional application prototype
Testing	1–2 weeks	Validated system
Deployment	1 week	Live application

13. ASSUMPTIONS

The following assumptions have been identified as factors that are considered to be true for the purpose of planning and developing the R-Soft system, although they have not yet been fully validated. These assumptions help define potential risks and dependencies that may impact the project's success.

• It is assumed that all intended users (players, team leaders, and field owners) have access to smartphones capable of running the application on either Android or iOS platforms.

• It is assumed that users will be willing to adopt the new system and transition from manual attendance tracking methods to a digital solution.

• It is assumed that internet connectivity, although unreliable during events, will be available intermittently to allow data synchronization after offline usage.

• It is assumed that free and open-source tools, frameworks, and services will be sufficient to support the development, deployment, and maintenance of the application within the given budget constraints.

• It is assumed that the QR code scanning functionality will perform reliably across a wide range of mobile devices and environmental conditions (e.g., outdoor lighting).

• It is assumed that the number of users and events will remain at a manageable level during the initial deployment phase, allowing the system to operate without significant performance issues.

• It is assumed that basic security measures implemented using free technologies will be sufficient to protect user data in the initial version of the system.
• It is assumed that testing conducted during real airsoft events will provide sufficient feedback to identify and resolve major system issues before full deployment.
• It is assumed that no major legal or regulatory requirements will impact the collection and storage of user data, including emergency contact information.

14. GLOSSARY

Term / Abbreviation	Explanation
QR Code	Scannable code used for check-in
Check-in	Recording attendance at an event
Offline Mode	Functionality without internet
Sync	Updating data when connection returns
Stakeholder	Person involved in the system

15. APPENDIX

The following wireframes illustrate the initial design of the R-Soft mobile application interface. These visuals provide an overview of key screens, including event management, attendance calendar and user data.

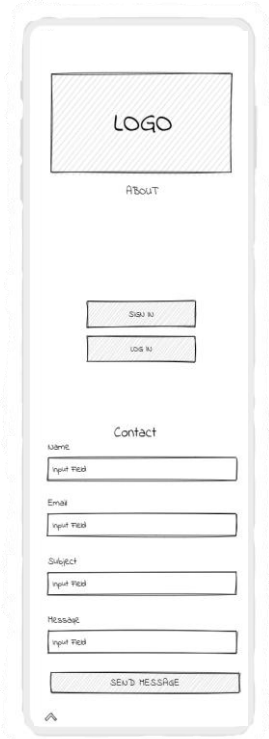
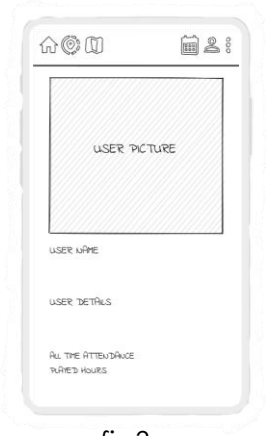


fig1

fig2



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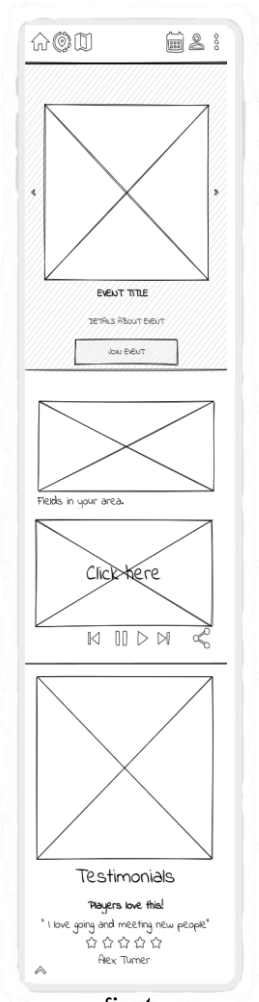

fig4

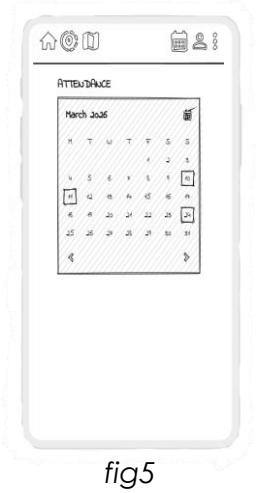

fig5

Figure fig1: Home Page
Figure fig2: Sign In Screen
Figure fig3: User Details Screen
Figure fig4: QR Check-in And Event Details Screen
Figure fig5: Attendance Calendar Screen